

1.5 The Lehmann representation and the spectral function

Källén-Lehmann representation: very useful for formal derivations

using set of eigenstates of full Hamiltonian $\mathbb{1} = \sum_n |n\rangle\langle n|$

disordered Green function: $G^>(v, t, t') = -i \langle c_v(t) c_v^\dagger(t') \rangle =$

$$= -i \frac{1}{Z} \sum_n \langle n | e^{-\beta H} c_v(t) c_v^\dagger(t') | n \rangle =$$

$$= -i \frac{1}{Z} \sum_{nn'} e^{-\beta E_n} \langle n | c_v | n' \rangle \langle n' | c_v^\dagger | n \rangle e^{i(E_n - E_{n'})(t - t')}$$

↓ Fourier transform

$$G^>(v, \omega) = -\frac{2\pi i}{Z} \sum_{nn'} e^{-\beta E_n} \langle n | c_v | n' \rangle \langle n' | c_v^\dagger | n \rangle \delta(E_n - E_{n'} - \omega)$$

analogously for $G^<(v, \omega) = -G^>(v, \omega) e^{-\beta \omega}$

$$G^R(v, \omega) = -i \int_0^\infty dt e^{i(\omega + i\eta)t} \frac{1}{Z} \sum_{nn'} e^{-\beta E_n} \left(\langle n | c_v | n' \rangle \langle n' | c_v^\dagger | n \rangle \right) \cdot e^{+i(E_n - E_{n'})t} =$$

$$= \frac{1}{Z} \sum_{nn'} e^{-\beta E_n} \left[\frac{\langle n | c_v | n' \rangle \langle n' | c_v^\dagger | n \rangle}{\omega + E_n - E_{n'} + i\eta} + \frac{\langle n | c_v^\dagger | n' \rangle \langle n' | c_v | n \rangle}{\omega - E_n + E_{n'} + i\eta} \right] =$$

$$G_R = \frac{1}{Z} \sum_{nn'} \frac{\langle n | c_v | n' \rangle \langle n' | c_v^\dagger | n \rangle}{\omega + E_n - E_{n'} + i\eta} \left(e^{-\beta E_n} + e^{-\beta E_{n'}} \right)$$

Lehmann representation

The Spectral function

We use $(\omega + i\eta)^{-1} = P \frac{1}{\omega} - i\pi \delta(\omega)$

$$\Rightarrow -\frac{1}{\pi} \ln G^R(\omega) = \frac{1}{\pi} \sum_{nn'} \langle n | c | n' \rangle \langle n' | c^\dagger | n \rangle (e^{-\beta E_n} + e^{-\beta E_{n'}}) \cdot \delta(\omega + E_n - E_{n'}) =$$

$$= \frac{1}{\pi} \sum_{nn'} |\langle n | c | n' \rangle|^2 (e^{-\beta E_n} + e^{-\beta E_{n'}}) \delta(\omega + E_n - E_{n'}) = A(\omega)$$

$\Rightarrow$ positive definite!

$\nearrow$
"Spectral function"

What is its significance?

•) $G^R(\omega) = \int d\omega' \frac{A(\omega')}{\omega - \omega' + i\eta}$ (comparison to Lehmann)

•) $\underline{G_0^R(\vec{k}, \omega)} = -i \int_{-\infty}^{\infty} dt \theta(t-t') e^{i\omega(t-t')} e^{-i\epsilon_{\vec{k}}(t-t')} \gamma(t-t') =$

$$= \frac{1}{\omega - \epsilon_{\vec{k}} + i\eta}$$

non-relativistic retarded Green function in band frequencies

$$\Rightarrow A_0(\vec{k}, \omega) = -\frac{1}{\pi} \ln G_0^R(\vec{k}, \omega) = \delta(\omega - \epsilon_{\vec{k}})$$

Spectral function is "density of states" for an on lattice system!

properties of the Spectral function

•) positive definite: $A(\nu, \omega) \geq 0$

•) obeys sum rule $\int_{-\infty}^{\infty} d\omega A(\nu, \omega) = 1$

•) average occupation of state ν :

$$\bar{n}_\nu = \langle c_\nu^\dagger c_\nu \rangle = -i G^<(\nu, t=0) = \\ = -i \int_{-\infty}^{\infty} d\omega G^<(\nu, \omega) = \int_{-\infty}^{\infty} d\omega A(\nu, \omega) n_F(\omega)$$

•) interactions broaden Spectral function

"wave packets" decay: $G^K(\nu, t) \approx -i\theta(t) e^{-i\epsilon t} e^{-t/\tau}$

$$\Rightarrow A(\nu, \omega) \approx \frac{1}{\pi} \frac{\frac{1}{\tau}}{(\omega - \epsilon)^2 + (\frac{1}{\tau})^2}$$

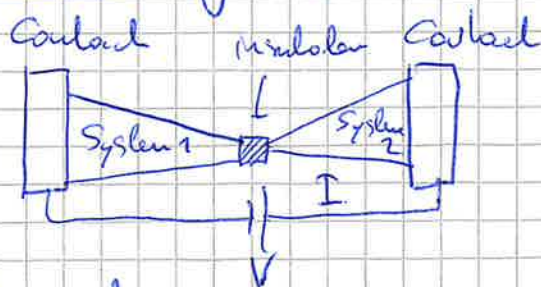
decay

Lorentzian broadening instead of sharp $\delta(\omega - \epsilon)$

Measuring the single-particle Spectral function

→ tunneling Spectroscopy

Schup:



scanning tunneling microscopy (STM)

System 1 and 2 are described by H_1 and H_2

with complete single-particle bases $\{| \nu \rangle\}$, $\{| \mu \rangle\}$
and operators $c_{1,\nu}^{(H)}$, $c_{2,\mu}^{(H)}$

Coupling of two sides: Amplitude overlap of wave functions

$$\Rightarrow H_T = \sum_{\nu\mu} (T_{\nu\mu} c_{1,\nu}^\dagger c_{2,\mu} + T_{\nu\mu}^* c_{2,\mu}^\dagger c_{1,\nu})$$

with $T_{\nu\mu} = \int d^3r \psi_\nu^*(\vec{r}) H(\vec{r}) \psi_\mu(\vec{r})$

Current through device: charge rate of particles $I_e = -e \langle \dot{I} \rangle$
with $I = \dot{N}_1$

$$I = i [H, N_1] = i [H_T, N_1] = e \sum_{\nu\mu} \sum_{\nu'} (T_{\nu\mu} c_{1,\nu}^\dagger c_{2,\mu} + T_{\nu\mu}^* c_{2,\mu}^\dagger c_{1,\nu}) c_{1,\nu'}^\dagger c_{1,\nu'}$$

$$\begin{aligned} \text{e.g.: } [c_{1,\nu}^\dagger c_{2,\mu}, c_{1,\nu'}^\dagger c_{1,\nu'}] &= \\ &= c_{1,\nu}^\dagger [c_{2,\mu}, c_{1,\nu'}^\dagger c_{1,\nu'}] + [c_{1,\nu}^\dagger, c_{1,\nu'}^\dagger c_{1,\nu'}] c_{2,\mu} = \\ &= c_{1,\nu}^\dagger (\{c_{2,\mu}, c_{1,\nu'}^\dagger\} c_{1,\nu'} - c_{1,\nu'}^\dagger \{c_{2,\mu}, c_{1,\nu'}\}) + \\ &\quad + (\{c_{1,\nu}^\dagger, c_{1,\nu'}^\dagger\} c_{1,\nu'} - c_{1,\nu'}^\dagger \{c_{1,\nu}^\dagger, c_{1,\nu'}\}) c_{2,\mu} = \\ &= -c_{1,\nu}^\dagger c_{2,\mu} \delta_{\nu\nu'} \end{aligned}$$

$$\Rightarrow I = -i \sum_{\nu\mu} (T_{\nu\mu} c_{1,\nu}^\dagger c_{2,\mu} - T_{\nu\mu}^* c_{2,\mu}^\dagger c_{1,\nu}) = -i (L \dot{\Phi}(t))$$

→ calculate particle current due to potential difference
(via H_T) in linear response theory

$$c(I)(t) = \int_{-\infty}^{\infty} dt' C_{I p H_T}^R(t, t')$$

$$C_{I p H_T}^R(t, t') = -i \theta(t-t') \langle [\hat{I}_p(t), \hat{H}_T(t')] \rangle_0$$

$$C_{\uparrow\uparrow}^R(t-t') = -\theta(t-t') \langle [\hat{L}(t), \hat{L}^\dagger(t'), \hat{L}(t') + \hat{L}^\dagger(t')] \rangle_0 =$$

$$= -\theta(t-t') \{ \langle [\hat{L}(t), \hat{L}(t')] \rangle_0 - \langle [\hat{L}^\dagger(t), \hat{L}^\dagger(t')] \rangle_0 + c.c. \}$$

The combination $\langle [\hat{L}(t), \hat{L}(t')] \rangle_0$ involves terms like

$$\langle c_{1,\nu}^\dagger(t) c_{2,\mu}(t) c_{1,\nu}^\dagger(t') c_{2,\mu}(t') \rangle_0$$

$\Rightarrow$ does not satisfy particle conservation! $\Rightarrow$ vanish (in each system)

$$\Rightarrow I_P(t) = 2 \operatorname{Re} \int_{-\infty}^{\infty} dt' \theta(t-t') \langle [\hat{L}^\dagger(t), \hat{L}(t')] \rangle_0 =$$

$$= 2 \operatorname{Re} \int_{-\infty}^{\infty} dt' \theta(t-t') \sum_{\nu\mu} \sum_{\nu'\mu'} T_{\nu\mu}^* T_{\nu'\mu'} \langle [\hat{c}_{2,\mu}^\dagger(t) \hat{c}_{1,\nu}(t), \hat{c}_{1,\nu'}^\dagger(t') \hat{c}_{2,\mu'}(t')] \rangle_0$$

$$= 2 \operatorname{Re} \int_{-\infty}^{\infty} dt' \theta(t-t') \sum_{\nu\mu} \sum_{\nu'\mu'} T_{\nu\mu}^* T_{\nu'\mu'} (\langle \hat{c}_{2,\mu}^\dagger(t) \hat{c}_{1,\nu}(t) \hat{c}_{1,\nu'}^\dagger(t') \hat{c}_{2,\mu'}(t') \rangle_0$$

$$- \langle \hat{c}_{1,\nu'}^\dagger(t') \hat{c}_{2,\mu'}(t') \hat{c}_{2,\mu}^\dagger(t) \hat{c}_{1,\nu}(t) \rangle_0)$$

For a non-interacting Hamiltonian we can use

Wick's theorem: $\langle c_1^{(+)} c_2^{(+)} c_3^{(+)} c_4^{(+)} \rangle_0 = \langle c_1^{(+)} c_2^{(+)} \rangle_0 \langle c_3^{(+)} c_4^{(+)} \rangle_0$

$$+ \langle c_1^{(+)} c_4^{(+)} \rangle_0 \langle c_2^{(+)} c_3^{(+)} \rangle_0$$

$$- \langle c_1^{(+)} c_3^{(+)} \rangle_0 \langle c_2^{(+)} c_4^{(+)} \rangle_0$$

$\uparrow$ odd permutations

$$\Rightarrow I_P(t) = 2 \operatorname{Re} \int_{-\infty}^{\infty} dt' \theta(t-t') \sum_{\nu\mu} \sum_{\nu'\mu'} T_{\nu\mu}^* T_{\nu'\mu'}$$

$$\cdot (\langle \hat{c}_{2,\mu}^\dagger(t) \hat{c}_{1,\nu}(t) \rangle_0 \langle \hat{c}_{1,\nu'}^\dagger(t') \hat{c}_{2,\mu'}(t') \rangle_0 + \langle \hat{c}_{2,\mu}^\dagger(t) \hat{c}_{2,\mu'}(t') \rangle_0 \langle \hat{c}_{1,\nu}^\dagger(t) \hat{c}_{1,\nu'}(t') \rangle_0$$

$$- \langle \hat{c}_{2,\mu}^\dagger(t) \hat{c}_{1,\nu'}^\dagger(t') \rangle_0 \langle \hat{c}_{1,\nu}(t) \hat{c}_{2,\mu'}(t') \rangle_0$$

$$- \langle \hat{c}_{1,\nu'}^\dagger(t') \hat{c}_{2,\mu'}(t') \rangle_0 \langle \hat{c}_{2,\mu}^\dagger(t) \hat{c}_{1,\nu}(t) \rangle_0 - \langle \hat{c}_{1,\nu'}^\dagger(t') \hat{c}_{2,\mu}^\dagger(t) \rangle_0 \langle \hat{c}_{2,\mu'}(t') \hat{c}_{1,\nu'}(t') \rangle_0$$

$$+ \langle \hat{c}_{1,\nu'}^\dagger(t') \hat{c}_{2,\mu'}^\dagger(t') \rangle_0 \langle \hat{c}_{2,\mu}(t) \hat{c}_{1,\nu}(t) \rangle_0)$$

$$= 2 \operatorname{Re} \int_{-\infty}^{\infty} dt' \theta(t-t') \sum_{\nu\mu} \sum_{\nu'\mu'} T_{\nu\mu}^* T_{\nu'\mu'}$$

$$(\langle \hat{c}_{2,\mu}^\dagger(t) \hat{c}_{2,\mu'}(t') \rangle_0 \langle \hat{c}_{1,\nu}(t) \hat{c}_{1,\nu'}^\dagger(t') \rangle_0$$

$$- \langle \hat{c}_{1,\nu'}^\dagger(t') \hat{c}_{1,\nu}(t) \hat{c}_{2,\mu'}(t') \hat{c}_{2,\mu}^\dagger(t) \rangle_0)$$

-) The applied voltage (difference in chemical potential) leads to a time-dependence. This can be pulled out:

$$\hat{c}_1(t) = \tilde{c}_1(t) e^{-i(-e)V_1 t}$$

$$\hat{c}_2(t) = \tilde{c}_2(t) e^{-i(-e)V_2 t}$$

-) Choose a basis of the decoupled system (w/o HT), where GF is diagonal: $G_{\mu\nu}^> = \delta_{\mu\nu} G_{\nu}^>$

-) change of variable: $t' \rightarrow t' + t$

$$I_p = 2ke \int_{-\infty}^0 dt' \sum_{\nu\mu} |T_{\nu\mu}|^2 e^{i(-e)(V_1 - V_2)t'}$$

$$[G_1^>(v_i, t) G_2^<(\mu_i, t') - G_1^<(v_i, t') G_2^>(\mu_i, t')]$$

Fowler-Nordheim ($V = V_1 - V_2$)

$$I_p = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \sum_{\nu\mu} |T_{\nu\mu}|^2 [G_1^>(v, \omega) G_2^<(\mu, \omega + eV) - G_1^<(v, \omega) G_2^>(\mu, \omega + eV)]$$

in terms of spectral functions

$$I_p = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \sum_{\nu\mu} |T_{\nu\mu}|^2 A_1(v, \omega) A_2(\mu, \omega + eV) [n_F(\omega + eV) - n_F(\omega)]$$

Sweep of external voltage $\frac{dI}{dV}$

We assume system 2 to be an ordinary metal: $\sum_{\mu} |T_{\nu\mu}|^2 A_2(\mu, \omega + eV) \approx \text{const.}$

$$\Rightarrow \frac{dI}{dV} \propto \int_{-\infty}^{\infty} d\omega \left[- \frac{\partial n_F(\omega + eV)}{\partial \omega} \right] \sum_{\nu} A_1(v, \omega)$$

$$\text{at low } T: \partial n_F \propto \delta \Rightarrow \left[\frac{dI}{dV} \propto \sum_{\nu} A_1(v, -eV) \right]$$

$\Rightarrow$ direct measurement of spectral function in STM!